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Full paper

Ultrasensitivity of self-powered wireless triboelectric vibration sensor for operating in underwater environment based on surface functionalization of rice husks

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ABSTRACT

We demonstrate the self-powered wireless triboelectric vibration sensor as made from the naturally nanoporous SiO_2 particles for allowing the detection of the vibrations and movement in the underwater environment. The nanoporous SiO_2 particles are directly converted from the rice husks (referred to as RH_{SiO2}), which exhibit strongly interacting surface hydroxyl groups. Through the enzymatic treatments, the surface potential of the RH_{SiO2} particles can be modulated to obtain either an extremely low or strongly high electronegativity. Specifically, by adding fluorinated groups using fluoroalkylsilane (FOTS) treatment to obtain RH_{SiO2}-F, the charge density of the RH_{SiO2}-F triboelectric nanogenerator (TENG) can be enhanced ~ 56.67-fold as compared to the untreated RH_{SiO2}-TENG. The power density of the RH_{SiO2}-F TENG is increased from 0.077 mWm^{-2} to 261 mWm^{-2}. The RH_{SiO2}-F particles are encapsulated in a quartz cube to fabricate a self-powered wireless sensor that can be stabilized for operating in an underwater environment at various temperatures. The theoretical calculation further demonstrates that the triboelectric potential is dramatically established between the surface functionalized RH_{SiO2}-F particles and the quartz's surface. With porous nature of rice husks covered with nano-Si is of a high functionality for designing a new-type TENG which has a great potential to apply in the environmental monitoring.

1. Introduction

The global effect of sensors have been of significant importance in the internet of things (IoT). The devices mounted with electronics, network connectivity, sensors, and actuators which enables these devices to exchange and transfer the data. Nowadays, because sensor nodes are generally battery-powered, which have a limited lifetime. Moreover, the rapid growth of materials and science technology yields miniaturized and stretchable electronics, with a various revolutionary technology on multifunctional devices. However, without the self-powered function, it is difficult to make the sensor to be deployed in harsh environments. Thus, considering the sustainable operation for most of the IoT devices, which have become challenges, especially for the inconvenience of having to replace batteries in devices such as environmental monitoring for high temperature and high humidity regions, deep ocean areas, the implanted biomedical sensor, and the complexly industry-treatment system.

Wang's group has invented the piezoelectric nanogenerator (PENG) [1–4] and the triboelectric nanogenerator (TENG) [5–8], that makes them a prodigious possibility to be a useful way for the future engineering applications to explore a variety of multifunctional energy harvester [9–12], self-powered devices [13–15], tribotronics [16–20], wearable electronics [21,22], and piezotronic devices [3,4,23,24]. However, one of the critical tasks for developing the high performance TENG is to create a highly specific surface area with a tunable electronegativity on both sides of the frication layers to obtain an enormously triboelectric potential [20,25,26]. Specifically, to find suitable solutions and materials are still the challenging tasks for solving

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those highly sensitive to the environment humidity of TENG that is vitally important to make commercialized products.

One of the most highly electronegativity material in the tribo-series table is the polytetrafluoroethylene (PTFE), of which, however, possesses the difficult processability, low surface area, and is cost expensive, limiting its feasible and stability for further developing to commercialize. In contrast, rice husks are an abundant agricultural byproduct, which possess rich eco-friendly silica composition with a variety of applications in materials science and has even become a common additive in food production. Our previous work discovered that the highly triboelectric potential of the rice husks were the superior candidate for designing a new-type TENG for lighting LED bulbs [27]. The three-dimensional (3D) porous nature of the rice husks covered with an enormous surface of pure nano-Si [27], which can easily become surface engineering to exhibit the strong bonds by the hydroxyl groups with the covalently linked trieoxysilane groups on their surfaces. Generally, the rice husks possess low electronegativity, and stabilize the positive charge, to have a strong tendency for repulsing the electron from its surface. Interestingly, can we take advantage of the 3D porous nature of the rice husks to modulate their electronegativity and create either a strongly positive charge or a negative charge on their surface? If so, the porous nature of the rice husks with these unique features such as low cost, lightweight, thermal and chemical stability, and environmental friendliness, etc., which could offer many distinct advantages to the nanotechnology community over the conventional chemical and/or physical synthesis method.

This work, the porous nature of the RH_{SiO2} particles, was directly derived from the rice husks through the acid hydrolysis and the thermal annealing process [27]. Through the enzymatically treated process, the RH_{SiO2} particles were immersed in the trieoxysilane solution using the different functional head groups such as $-NH_2$, $-Cl$, and $-F$ [28,29], the terminated bonds $-NH_2$, $-Cl$, and $-F$ were therefore chemically bonded on the surface of the RH_{SiO2} particles, respectively. After enzymatically treated process, the charge density of the RH_{SiO2}-F TENG was significantly enhanced ~ 56.67-fold in contrast to that of the untreated RH_{SiO2} TENG. Furthermore, the surface-functionalized RH_{SiO2}-F particles were successfully fabricated into a self-powered wireless vibration sensor, which can be directly monitoring the mechanical motion in an underwater environment. Considering the sensitive to humidity of TENG, this is the first report that the TENG can be stabilized for operating in a water environment at varying temperatures that will further deliver a significant progress to be highly possible for becoming a commercialized product.

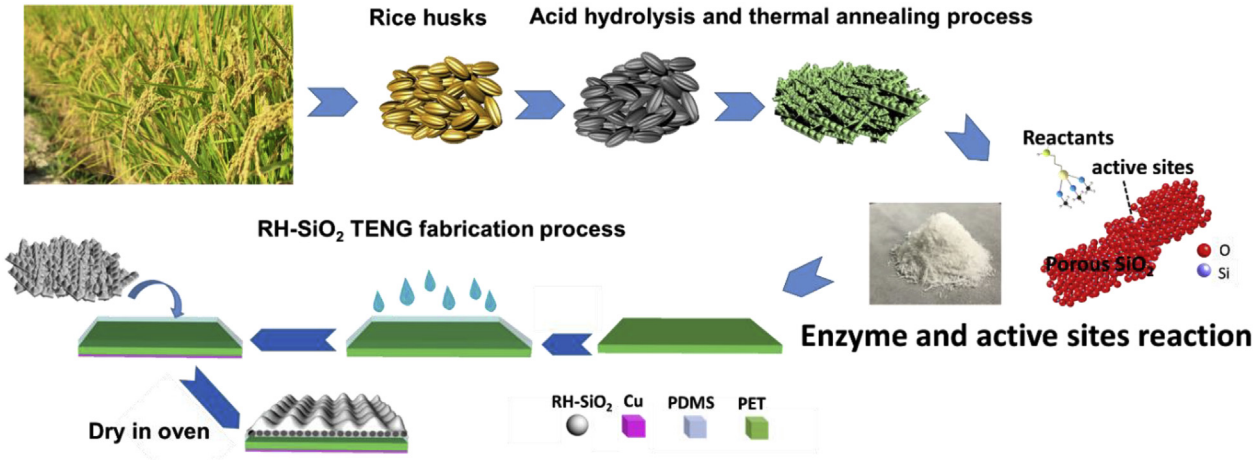
2. Experimental section

The preparation of the porous RH-SiO_2 particles, immobilization of the enzyme, and TENG's fabrication process, are shown in the schematic diagram of Scheme 1. First, the nanoporous rice husk SiO_2 (RH_{SiO2}) particles were prepared through the acid hydrolysis and thermal annealing process [27]. Next, the RH_{SiO2} particles were washed by deionized water and ethanol solution for several times to remove the impurities and collected by the centrifugation process for 10 min. Then, the surface-functionalized RH_{SiO2} particles were dried in an oven at 60 °C for 24 h. After that, to proceed the silanized nanoporous RH-SiO_2 particles by enzymatically treated process, the RH_{SiO2} particles were immersed in the precursors of the (3-aminopropyl)triethoxysilane (APTES), (3-chloropropyl)trimethoxysilane (CPTMS), and 1H, 1H, 2H, 2H-perfluorooctyltriethoxysilane (FOTS), as referred to by the RH_{SiO2}-NH_2, RH_{SiO2}-Cl, and RH_{SiO2}-F, respectively (so-called the surface-functionalized or enzymatically treated process). The RH_{SiO2} particles (untreated) were served as a friction layer while the RH_{SiO2}-NH_2, RH_{SiO2}-Cl, and RH_{SiO2}-F were individually used as another friction layer, respectively. To fabricate the rice husk TENG using the surface-functionalized RH_{SiO2} particles, first, polydimethylsiloxane (PDMS) elastomer and cross-linker were uniformly mixed in a 10:1 ratio (w/w) and degassed for 30 min. Next, the PDMS was spin-coated (at 1000 rpm) onto a Cu-coated PET film. Then, the RH_{SiO2}, RH_{SiO2}-NH_2, RH_{SiO2}-Cl, and RH_{SiO2}-F particles were individually transferred onto the surface of the PDMS film and dried at 80 °C for 1 h, leading to the RH_{SiO2} particles that were tightly bonded on the surface of the PDMS film to form the RH_{SiO2}-TENG devices ($3 \times 3\ cm^2$). The surface's morphologies of the RH_{SiO2} particles were investigated by the field emission scanning electron microscope (FESEM, HITACHI S4800). The surface-functionalized RH_{SiO2} particles were confirmed by the high-resolution X-ray photoelectron spectrometer (UL VAC-PHI XPS) and a contact angle meter (Model 100SB). The RH_{SiO2}-F particles were encapsulated in the quartz cube to fabricate a vibration sensor, which was evaluated in water and/or in ambient air by employing a programmable constant temperature and humidity chamber (Thermotron SE-1000-3-3) to investigate their electrical output performance. The output short-circuit current and open-circuit voltage of the TENG were measured using a low-noise current preamplifier (Stanford Research System Model SR570), a low-noise voltage preamplifier (Stanford Research System Model SR560), and a Keithley 6514 programmable electrometer. The TENG was also tested in an underwater environment at various temperatures to evaluate its electrical output performance.

3. Results and discussion

Fig. 1a reveals that the regular bumps morphology was observed on the surface of the RH_{SiO2} particles, which was directly converted from the rice husks as shown in the inset image of Fig. 1a. The cross sectional image of Fig. 1b shows that the interconnected nanoporous structure naturally exists in the rice husks. Furthermore, the XPS of O1s spectra in Fig. 1c evidences that the RH_{SiO2} particles exhibit the abundant hydroxyl groups (OH) and the Si—O bonds at 531.7eV and 532.7 eV, respectively [30]. The surface contact angle of the RH_{SiO2} particles has ~ 65.5°, as shown in Fig. 1d. Through the enzymatically treated process, the RH_{SiO2} particles were surface-functionalized by grafting end-amine (APTES), end-chlorine (CPTMS), and end-fluorine (FOTS), the corresponding contact angle of the surface treated RH_{SiO2} $-NH_2$, RH_{SiO2} $-Cl$, and RH_{SiO2} $-F$ particles were measured at 40.3°, 112.8°, and 143.2°, as shown in Fig. 1e–g, respectively. The characteristic RH_{SiO2}'s surface was significantly transformed from the hydrophilic to the hydrophobic surface. Furthermore, the surface-functionalized RH_{SiO2}-NH_2 particles in Fig. 1h reveals that the N 1s peak at 400 eV can further be deconvoluted into 399.62 eV and 401.75 eV, which were attributed to the free- and protonated amine groups [31], respectively. Fig. 1i–j reveal that the Cl 2p peak and F1s peak at 200 eV and 688.8 eV, respectively, evidence that the $-Cl$, and $-F$ have been intercalated onto the surface of the RH_{SiO2} particles [29,32,33].

The Kelvin probe force microscope (KPFM) was employed to quantitatively determine the work functions for the several surfaces treatment of the particles such as the RH_{SiO2}, RH_{SiO2}-NH_2, RH_{SiO2}-Cl, and RH_{SiO2}-F (see detail in Supporting Information of Fig. S1). Noticeably, Fig. 2a further reveals that the RH_{SiO2}-F particles possess the lowest surface potential with the highest value of work function among all the samples. To understand the TENG's output performance, the surface treated RH_{SiO2} particles were used as a bottom friction layer while the untreated RH_{SiO2} particles were utilized as a top friction electrode to fabricate the surface treated RH-$_{SiO2}$ TENG, as shown in the inset image of Fig. 2b. The corresponding open-circuit voltage and short-circuit current of the RH_{SiO2} (untreated) TENG, RH_{SiO2}-NH_2 TENG, RH_{SiO2}-Cl TENG, and RH_{SiO2}-F TENG are as shown in Fig. 2b–e, respectively. As expected, the voltage increases by increasing the loading resistance due to the Ohmic loss, while the current decreased by increasing the loading resistance for all the TENGs. Fig. 2b reveals that the output voltage and current of the untreated RH_{SiO2}-TENG are measured at 3 V and 6×10^{-8} A, respectively. Noted that the extremely small output power was ascribed to the surface roughness and size differences on both of the tribo-electrification layers while



Scheme 1. Schematic representation of the fabrication process of the TENG and immobilization of the enzyme using porous RH_{SiO2} particles.

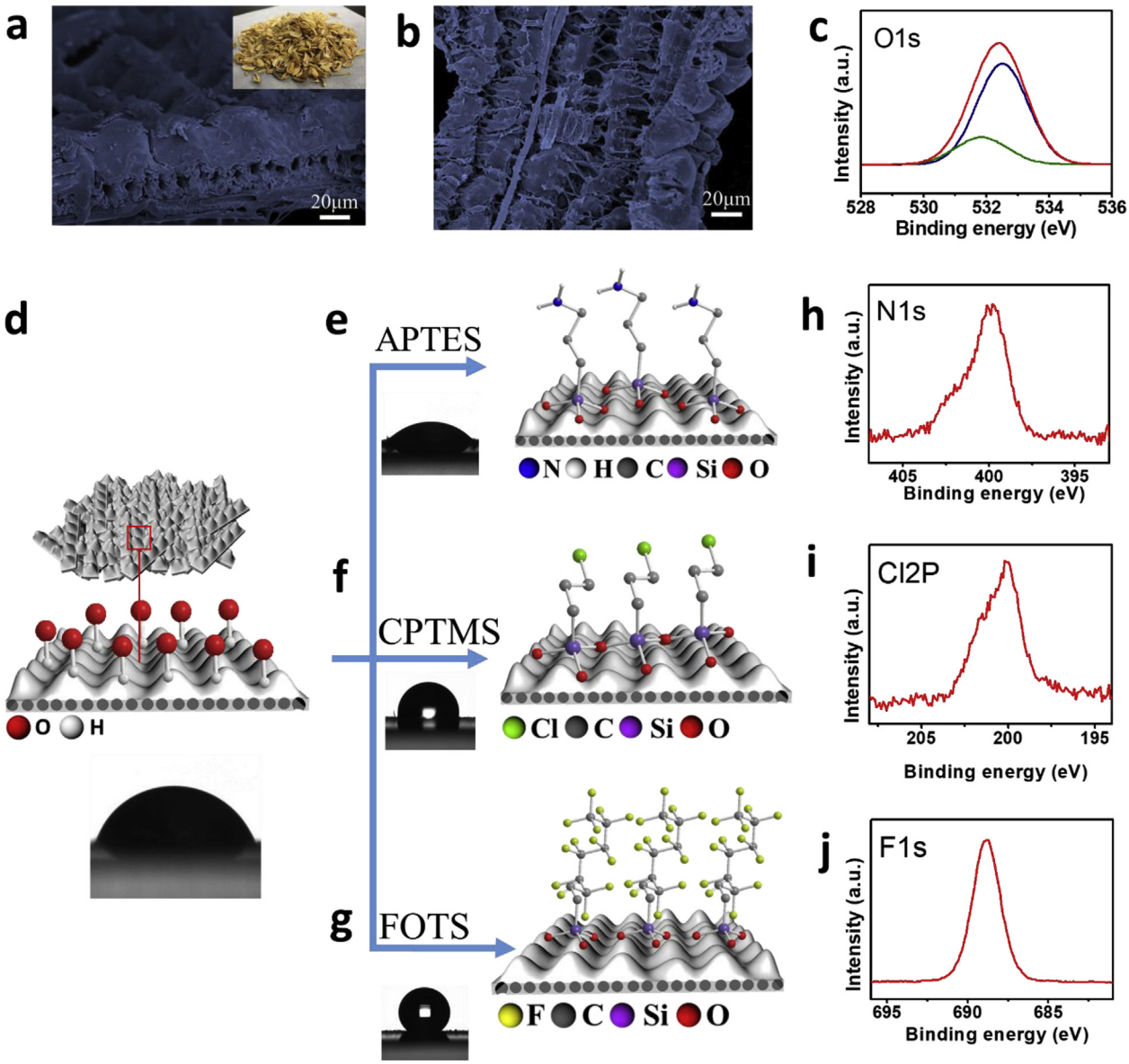


Fig. 1. (a) The SEM image of the RH_{SiO2} particles and the digital photographic of rice is inset. (b) The cross sectional image of the rice husks. (c) The XPS spectrum of O1s for the pristine RH_{SiO2}. (d) The schematic illustration of the pristine RH_{SiO2} and its contact angle measured at ~ 65.5°. (e) The schematic illustration of surface functionalization of RH_{SiO2} by APTES treatment and its contact angle at 40.3°, (f) the CPTMS treatment with its contact angle at ~ 112.8°, and (g) the FOTS treatment with its contact angle at ~143.2°, (h)–(j) the corresponding XPS spectra of N1s, Cl2p, and F1s, respectively.

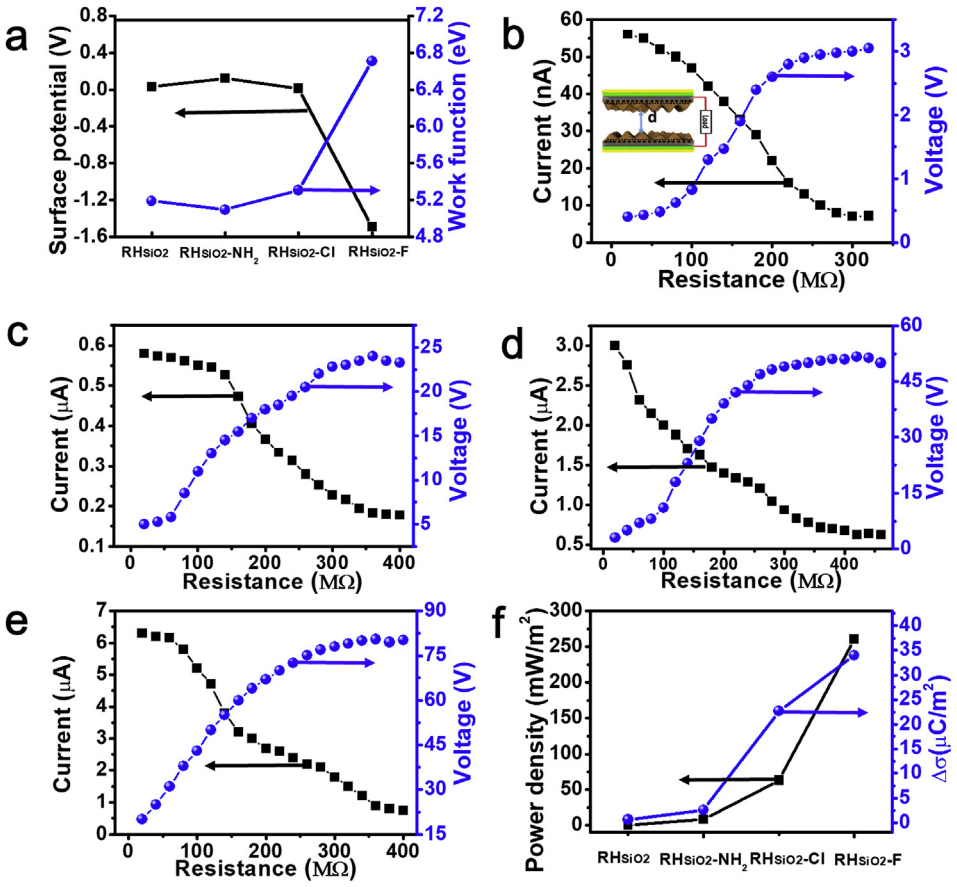


Fig. 2. (a)The work function and surface potential for the enzymatically treated process of the RH_{SiO2}–NH_2, RH_{SiO2}–Cl, and RH_{SiO2}–F, and pristine RH_{SiO2} particles, and the dependence of output current and voltage on the loading resistances for the different surface treatment of (b) the pristine RH_{SiO2} particles, the configuration of the TENG is shown in inset image, (c) the RH_{SiO2}-NH_2, (d) RH_{SiO2}-Cl, (e) RH_{SiO2}-F, and (f) The corresponding output power density and charge density for the different surface treatments of RH_{SiO2} particles.

preparing the top and bottom electrodes, although the untreated RH_{SiO2}-TENG was theoretically not expected to exhibit any electrical output [34]. After the APTES treatment, Fig. 2c reveals that the voltage and current of the RH_{SiO2}-NH_2 TENG are measured at 23 V and 6×10^{-7} A, respectively. The output current is almost increased by one order of magnitude as compared with the untreated RH_{SiO2}-TENG. After the CPTMS treatment, the voltage and current of the RH_{SiO2}-Cl TENG were measured at 50 V and 3×10^{-6} A, respectively, as shown in Fig. 2d. Importantly, after the FOTS treatment, the voltage and current of the RH_{SiO2}-F TENG were obtained at ~80 V and 6×10^{-6} A, respectively (see Fig. 2e), which is the highest value among them. Noted that all the TENGs have almost shown the same loading resistance around $1.2–2.6 \times 10^8$ Ω. Fig. 2f further reveals that the charge density of the RH_{SiO2}, RH_{SiO2}-NH_2, RH_{SiO2}-Cl, and RH_{SiO2}-F are measured at 0.6, 2.58, 22.7, and 34 mC/m^3, with the corresponding power density of 0.077, 8.5, 63.2, and 261 W/m^3, respectively. Through the FOTS treatment, the power density of the RH_{SiO2}-F TENG reaches ~ 261 mW/m^2, which is 3600-fold higher than that of the untreated RH_{SiO2} TENG.

The theoretical calculation was conducted to evaluate as to how the triboelectric potential for all the enzymatically treated the RH_{SiO2}-TENGs. For all the corresponding simulated TENG's configurations are as shown in the inset images of Fig. 3a–d. The surface treated RH_{SiO2} film was used as the bottom friction layer while the untreated RH_{SiO2} film acted as a top friction electrode, respectively. Obviously, Fig. 3a shows that the triboelectric potential is very low for both the top and bottom electrodes of the untreated RH_{SiO2} films. However, after the enzymatic treatment by grafting end-amine (APTES), end-chlorine (CPTMS), and end-fluorine (FOTS) on the porous RH_{SiO2}'s surface, Fig. 3b shows that the triboelectric potential of the RH_{SiO2}-NH_2 film reaches ~ 53 V, which is four times increased in contrast to that of the untreated RH_{SiO2} film. This is because the $-NH_2$ groups exhibit the highly positive surface potential, which strongly repel the electron from its surface to the untreated RH_{SiO2} film (see inset Fig. 3b). Therefore, when the untreated RH_{SiO2} film was contacted with the RH_{SiO2} $-NH_2$ film, the formation of the triboelectric potential was created between the two films. In contrast, after the nanoporous RH_{SiO2} particles were enzymatic treated by the CPTMS to expose the −Cl bonds on the RH_{SiO2}'s surface, The inset Fig. 3c displays that the RH_{SiO2}-Cl film was strongly attract the electron from the untreated RH_{SiO2}'s surface, and created a highly negative triboelectric potential (~ −331 V) on the surface of the RH_{SiO2}-Cl film, as shown in Fig. 3c. Specifically, Fig. 3d further reveals that the RH_{SiO2} particles were treated by FOTS process, which can create an enormous triboelectric potential between the surface of the untreated RH_{SiO2} and the RH_{SiO}-F film. The electrical potential distribution of the FOTS treatment is 53-fold higher than that of the untreated RH_{SiO2} film. Therefore, as shown in Fig. 4a–c, the RH_{SiO2} particles possess the abundant hydroxyls bonds (OH bonds), which were reacted with C−H bonds to graft onto the RH_{SiO2} particles' surface through the enzymatically treatment of the APTES, CPTMS, and FOTS process to form the terminal sites of the $-NH_2$, −Cl, and −F groups on such enzymes' surface, respectively. This explains how a superhydrophobic surface with a low work function was observed in the RH_{SiO2}'s surface, after being treated by the FOTS process (see Fig. 1j above). Therefore, through the enzymatically treated process, the surface charge of the naturally-inherited 3D nanoporous morphology of the RH_{SiO2} particles can be modulated to either become a very positive charge or an extremely negative charge when compared to its original surface charge, as summarized in Fig. 4d.

To overcome the TENG's output performance being affected by humidity, the functionalized RH_{SiO2}-F particles were encapsulated in a quartz cube to fabricate a self-powered vibration sensor, as shown in Fig. 5a. The self-powered vibration sensor was connected on the linear motor to simulate the mechanical-force-induced vibrations. The corresponding output voltage, current, and charge density of the sensor were

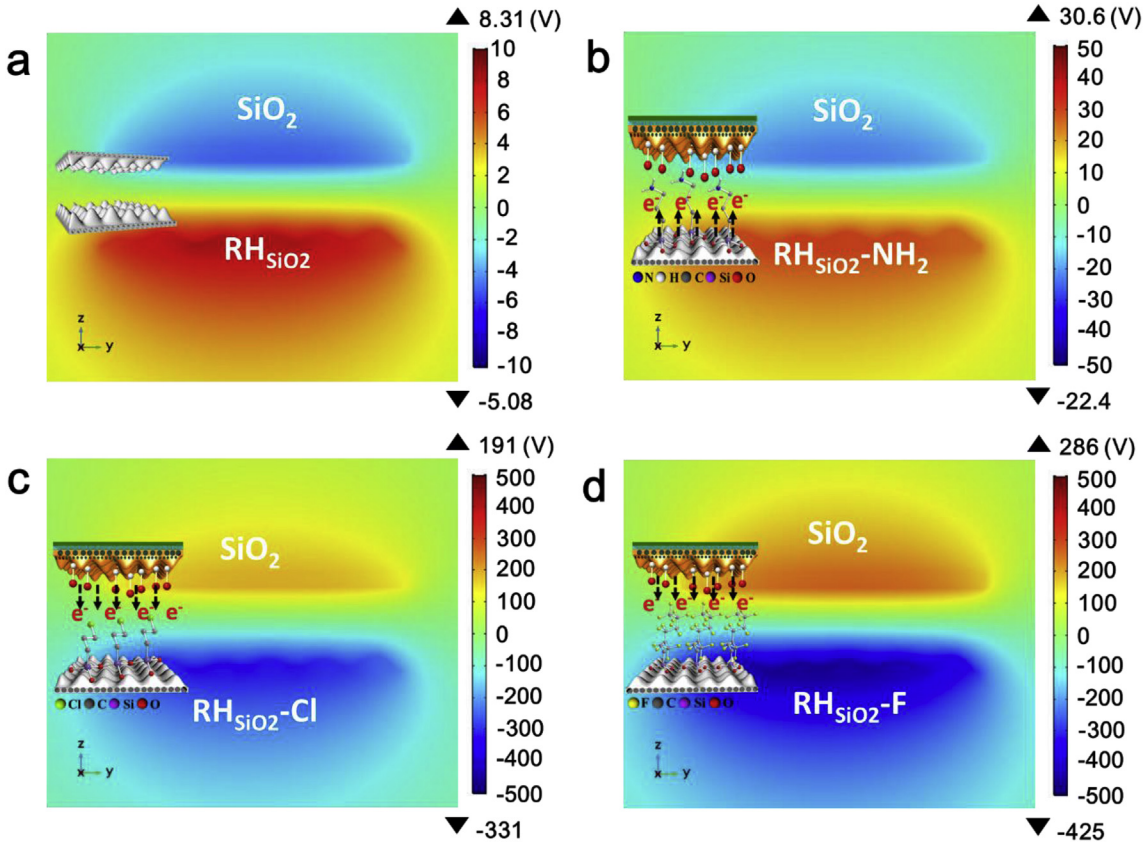


Fig. 3. The theoretical calculation of the triboelectric potential distribution for (a) untreated RH_{SiO2} film, and the enzymatically treated RH_{SiO2}-TENGs by grafting (b) end-amine (APTES), (c) end-chlorine (CPTMS), and (d) end-fluorine (FOTS).

measured at ~ 0.6 V, 40 nA, and 0.3 nC, as shown in Fig. 5b–d, respectively. Fig. 5e is the output voltage for each vibration state, which is corresponding to Fig. 5f–j, respectively. To explain the working mechanism, first, when the quartz cube was placed at a static state, the RH_{SiO2}-F particles were positioned in the bottom side of the quartz cube to establish a charge neutrality, as shown in Fig. 5f. The corresponding electrical output is zero, as shown in Fig. 5e (state A). Next, when the quartz cube was instantaneously mechanically vibrated, the RH_{SiO2}-F

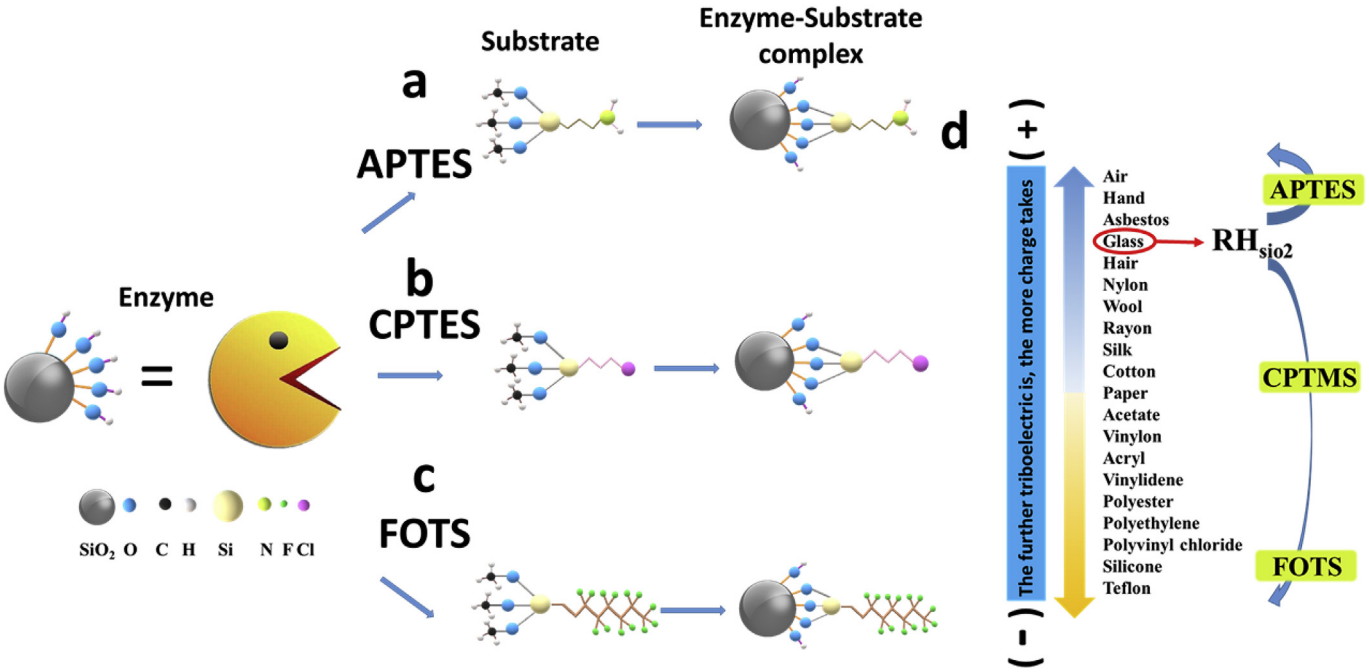


Fig. 4. (a)–(c) The enzymatically treated process through (a) APTES, (b) CPTMS, and (c) FOTS. (d) The summary of triboelectric potential table for the different substances.

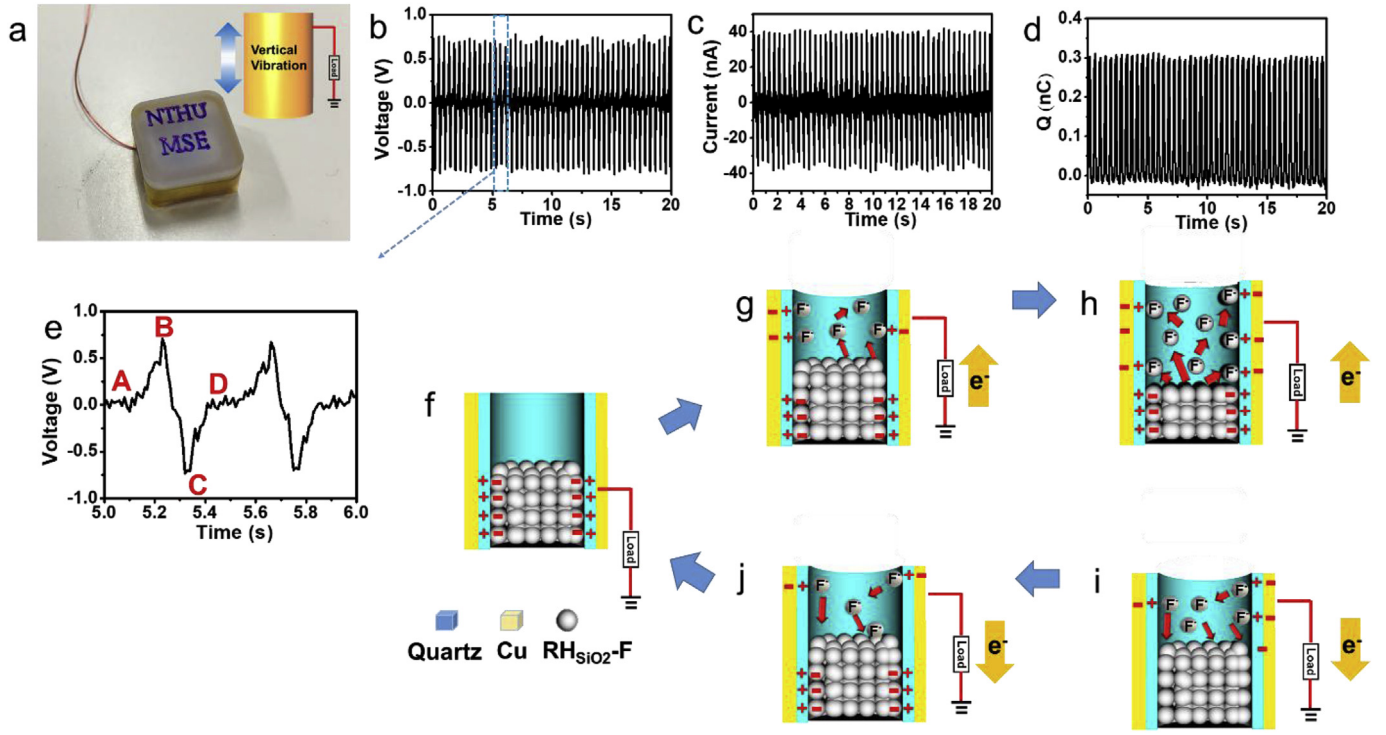


Fig. 5. (a) The digital photographic of the self-powered vibration sensor encapsulated by RH_{SiO2}-F particles, the corresponding output electrical signal of (b) voltage, (c) current, (d) charge density, (e) the experimentally measured output voltage from the self-powered variation sensor, showing the four different stages described in the following steps of the sensor in (f) static state, (g)–(h) vertical vibration (shaken up), (i)–(j) recovery to a static state (shaken down).

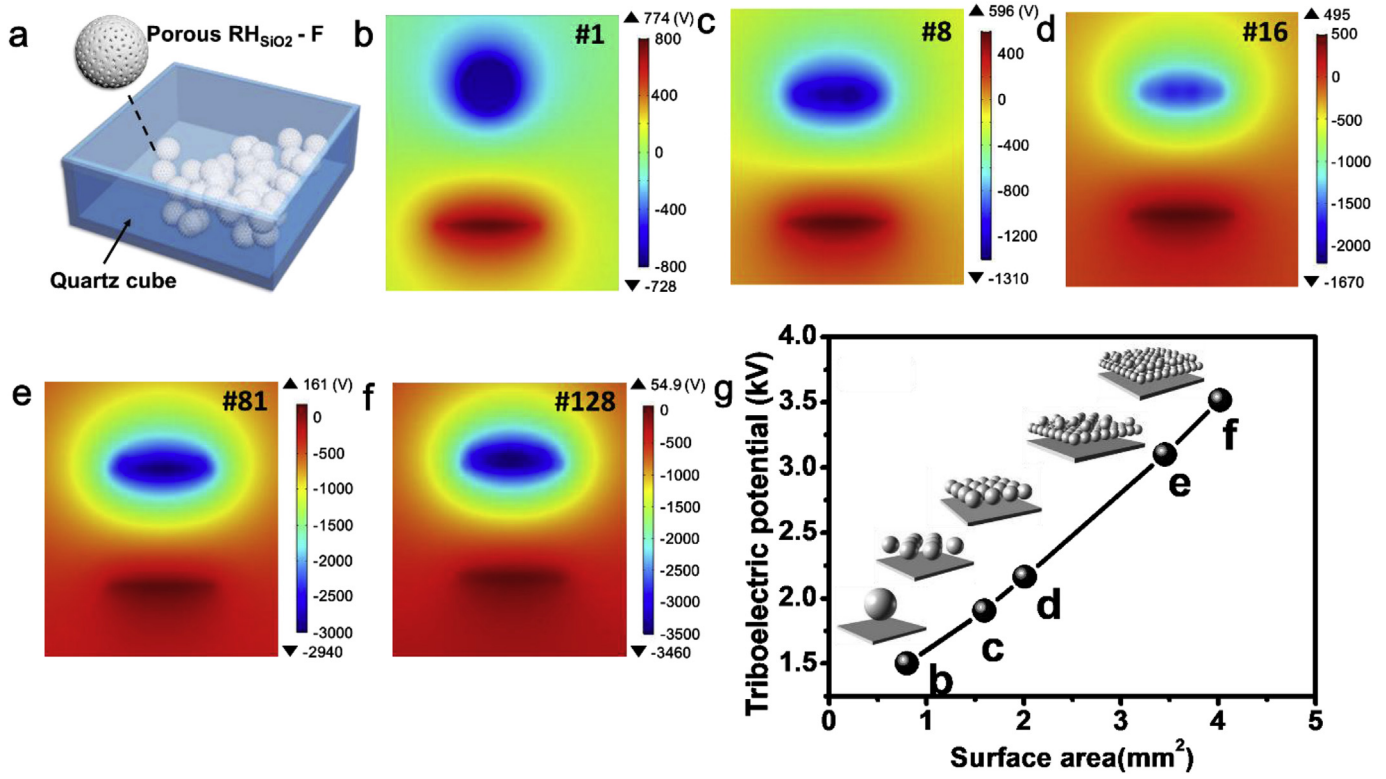


Fig. 6. (a) Schematic diagram showing the RH_{SiO2}-F particles encapsulated in a quartz cube, (b)–(f) The corresponding theoretical calculation of the triboelectric potential between the RH_{SiO2}-F particles and quartz's surface for the difference of particle numbers of #1, #8, #16, #81, #128, (g) The total triboelectric potential as a function of corresponding particle numbers and surface areas.

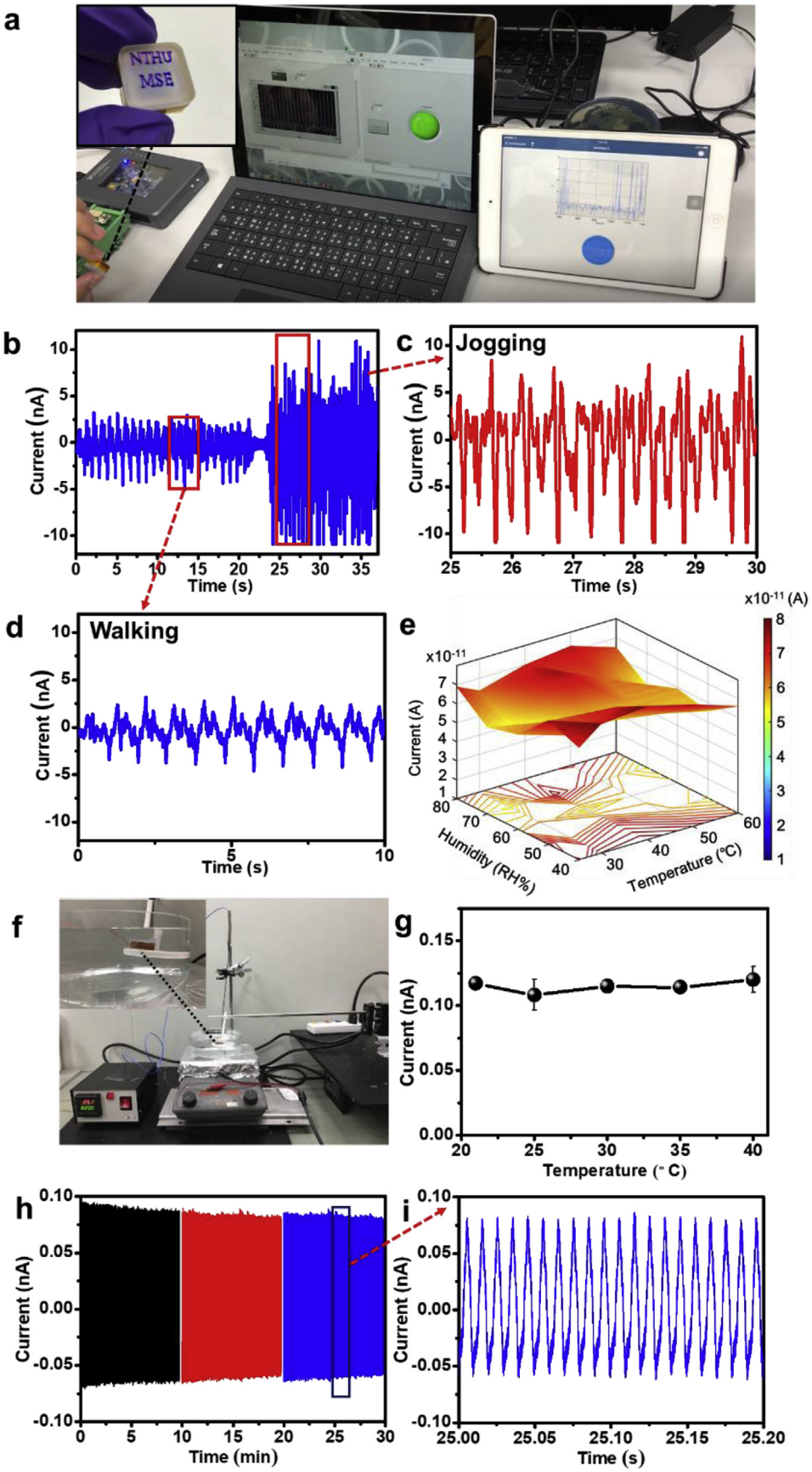


Fig. 7. (a) The photographic of the self-powered wireless vibration sensor, simultaneously detect the motion by a laptop and smart pad. (b)–(d) The current output for different human motions. (e) The output electricity versus the variation of the relative humidity and temperature. (f) The photographic image shows that the wireless vibration sensor was tested in water. (g) The sensors' current output versus the variations of the water temperature. (h)–(i) The stability test of the sensor at 25 °C for 30 min in water environment.

particles were shaken up and down to contact with the inner quartz's surface. The contacted areas between the RH_{SiO2}-F particles and the quartz's surface were increased, and the induced electrons therefore flowed from the ground state to accumulate on the electrode and produced the electricity (see Fig. 5g). When the contacted area between the RH_{SiO2}-F particles and the quartz's surface reached the largest areas (see Fig. 5h), which was corresponding to produce a maximal output voltage (see Fig. 5e, peak B). After that, when the RH_{SiO2}-F particles were separated from the quartz's surface and fell to the bottom side of the quartz cube, the contact areas between the RH_{SiO2}-F particles and the quartz cube's inner surface was reduced to produce the negative charge (see Fig. 5i). The corresponding electrical output signal was therefore transiting from peak B to peak C, as shown in Fig. 5e, until the accumulated electrons flowed back from the copper electrode to the ground state. Finally, the output current was progressively decreased to zero when most of the particles fell to the quartz cube's bottom side, as shown in Fig. 5j. The corresponding output signal was recovered to zero, as shown in Fig. 5e (D state). Noted that the signal shapes are strongly dependent on the vibrated direction, which may further be applied for detection of the different vibration modes and movement-tracking applications.

The schematic diagram in Fig. 6a shows the porous RH_{SiO2}-F particles were encapsulated in a quartz cube. The average particle's size distribution is around 8.8 μm (see Supporting Information of Fig. S2). The theoretical calculation was further demonstrated using the COMSOL software package [35,36] to understand the triboelectric potential's distribution of the rice husks (i.e., RH_{SiO2}-F particles) in a quartz cube. The calculation is based on the same volume with the different particle size and number for each theoretical model. When the particle number was increased from #1 to #128, the corresponding triboelectric potential are shown in Fig. 6b–f. Noticeably, because the electrostatic induction of the tribo-charges being formed on the quartz's surface and the RH_{SiO2}-F particles, a positive triboelectric potential was created on the bottom electrode (eg., quartz surface) while a negative potential (eg., RH_{SiO2}-F surface) was developed on the top electrode. Therefore, the smaller the particle size, the larger the surface area, an significantly negative triboelectric potential is observed. The fact further implies that the 3D morphology of porous rice husks with a high surface area could offer an enormously triboelectric potential to the quartz's surface by adopting the surface functionalized treatment. Fig. 6g further displays that the total triboelectric potential difference as a function of the surface areas and the number of particles, revealing that the higher the surface areas, the higher the triboelectric potential is obtained.

The vibration sensor encapsulated the RH_{SiO2}-F particles in a quartz cube was wireless integrated into a portable smart pad to simulate the human activity, as shown in Fig. 7a. The live videos (see Supporting Information, video-I and video-II) show that as the quartz cube was vibrated, the output signal can be clearly distinguished and counted. The live video-III (see Supporting Information) is showing the sensor for operating in water environment. Fig. 7b–d reveal that the sensor can further distinguish the human activities during walking and/or jogging, which corresponds to the output current of ~ 2.5 nA and 10 nA, respectively. The RH_{SiO2}-F particles have completely been encapsulated in the quartz to isolate the particles from the moisture problem. The performance of the vibration sensor was evaluated by employing a programmable constant temperature and humidity chamber. As shown in Fig. 7e, the vibration sensor shows the very stable output signals for both the plot of step change in relative humidity from 40% to 80% RH and varied the temperature from 25 °C to 80 °C. More importantly, the stability test of the sensor was proceeded in water at various temperatures to simulate the critical environments, as shown in Fig. 7f. When the water's temperature was varied in the range of 25 °C - 40 °C, the sensor still shows a very stable output signals, as shown in Fig. 7g. Even though the sensor was operated in water at 25 °C for 30 min, it displayed a highly sensitive and stabile for the mechanical vibration as compared to the same device being operated in air (see Fig. 7h and i). This is the first report that the TENG can be stabilized to operate in an underwater environment at varying temperatures. By integrating the environmental aspects into the eco-design product has a high potential applications to implement the materials for commercial products.

4. Conclusion

In summary, this is the first report to demonstrate the self-powered wireless triboelectric vibration sensor as made from the naturally porous rice husks, which can directly operate in an underwater environment at various temperatures. The RH_{SiO2} particles exhibit the highly dense Si−OH and OH bonds which are covalently linked with the trieoxysilane groups. Through the enzymatically treated process using the head groups of the $-NH_2$, −Cl, and −F, the terminated bonds $-NH_2$, −Cl, and −F were chemically formed on the surface of the RH_{SiO2} particles, respectively. Therefore, the electronegativity of the 3D naturally porous rice husks can be modulated to form either a very positive or extremely negative charge in contrast to its original surface charge. As a result, the charge density of the RH_{SiO2}-F TENG was significantly enhanced ~ 56.67 times in contrast to the untreated RH_{SiO2} TENG. After the FOTS treatment, the power density of the triboelectric nanogenerator (RH_{SiO2}-F TENG) was increased from 0.077 mWm^{-2} to 261 mWm^{-2}, which was enormously enhanced to ~ 339000%. The theoretical calculation further supports that the induced charge separation has been observed across the RH_{SiO2} and quartz's surface. The rice husks possess these abundantly unique features, such as being lightweight, thermal, and chemical stability, biodegradable, and environmental friendliness. Simply stated the porous nature covered with nano-Si is of a high functionality for designing a new-type triboelectric nanogenerator in the greater potential for detection of the vibrations and movement-tracking applications in an underwater environment.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nanoen.2019.03.067.

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